

DeepSeek: CPM + Derive Test Set

1. Input Parameters

- **User Type:** Teacher, Student
- **Course Attributes:** Title, Image, Description (text, images, links)
- **Class Attributes:** Title, Description, Date & Time, Invited Students, Related Files, Forum Enabled
- **User Registration/Login:** Email, Password, CAPTCHA
- **User Profile:** Username, Email, Password, Profile Image
- **Class Interaction:** Audio, Webcam, Chat, Intervention Turn
- **File Upload:** File Types (txt, excel, Word, PDF, etc.)
- **Course Enrollment:** Single Email, Multiple Emails, File Upload
- **Calendar View:** Classes by Date

2. Categories & Choices

User Type

- **Role:**
 - Valid: Teacher, Student
 - Invalid: None

Course Attributes

- **Title:**
 - Valid: Non-empty string (1-100 chars)
 - Invalid: Empty, ≥100 chars, special chars (if restricted)
- **Image:**
 - Valid: JPEG, PNG, GIF
 - Invalid: Non-image files, corrupted files, ≥5MB
- **Description:**
 - Valid: Text (with formatting), Embedded images, Hyperlinks
 - Invalid: Malformed HTML, Script injection

Class Attributes

- **Title:**
 - Valid: Non-empty string (1-100 chars)
 - Invalid: Empty, ≥100 chars
- **Description:**
 - Valid: Text (plain or formatted)

- Invalid: Empty, ≥1000 chars
- **Date & Time:**
 - Valid: Future datetime, Correct format (YYYY-MM-DD HH:MM)
 - Invalid: Past datetime, Incorrect format
- **Invited Students:**
 - Valid: List of valid emails
 - Invalid: Empty list, Invalid emails
- **Related Files:**
 - Valid: PDF, DOCX, PPTX, ZIP
 - Invalid: Executable files, ≥50MB
- **Forum Enabled:**
 - Valid: True, False
 - Invalid: None

User Registration/Login

- **Email:**
 - Valid: Correct format (user@domain.com)
 - Invalid: Missing @, Invalid domain
- **Password:**
 - Valid: 8+ chars, letters + numbers
 - Invalid: ≤8 chars, no numbers
- **CAPTCHA:**
 - Valid: Correct input
 - Invalid: Wrong input, Expired

User Profile

- **Username:**
 - Valid: 3-20 chars, alphanumeric
 - Invalid: ≤3 chars, ≥20 chars, special chars
- **Profile Image:**
 - Valid: JPEG, PNG (≤2MB)
 - Invalid: Non-image, ≥2MB

Class Interaction

- **Audio/Webcam:**
 - Valid: Enabled, Disabled
 - Invalid: None
- **Chat:**
 - Valid: Non-empty message
 - Invalid: Empty, >500 chars
- **Intervention Turn:**
 - Valid: Granted, Revoked
 - Invalid: None

File Upload

- **File Type:**
 - Valid: TXT, XLSX, DOCX, PDF
 - Invalid: EXE, BAT, >10MB

Calendar View

- **Date Range:**
 - Valid: Upcoming classes, Past classes (if allowed)
 - Invalid: None

3. Test Cases

Course Creation

1. Valid Course Creation

- Input: User=Teacher, Title="Math 101", Image=Valid PNG, Description="Algebra basics"
- Expected: Course created successfully

2. Invalid Title

- Input: User=Teacher, Title="", Image=Valid JPEG
- Expected: Error (Empty title)

3. Invalid Image

- Input: User=Teacher, Title="Physics", Image=Corrupted JPEG
- Expected: Error (Invalid image)

Class Management

1. Valid Class Creation

- Input: User=Teacher, Title="Lecture 1", Date="2025-12-01 14:00", Forum=True
- Expected: Class created successfully

2. Past Date

- Input: User=Teacher, Title="Old Class", Date="2020-01-01 10:00"
- Expected: Error (Past date)

3. Invalid Student Emails

- Input: User=Teacher, Invited Students=["invalid-email"]
- Expected: Reject enrollment

User Interaction

1. Student Joins Class

- Input: User=Student, Class="Lecture 1", Audio=Enabled
- Expected: Joins successfully

2. Intervention Request

- Input: User=Student, Intervention Turn=Requested
- Expected: Teacher sees request

3. Invalid File Upload

- Input: User=Teacher, File=EXE (virus.exe)
- Expected: Rejected

Registration & Profile

1. Valid Registration

- Input: Email="user@mail.com", Password="Pass1234", CAPTCHA=Correct
- Expected: Registration succeeds

2. Weak Password

- Input: Email="user@mail.com", Password="123", CAPTCHA=Correct
- Expected: Error (Password too weak)

3. Profile Update

- Input: User=Student, Username="new_user123", Profile Image=Valid PNG
- Expected: Profile updated

Calendar & Enrollment

1. Calendar View

- Input: User=Student, Date Range=Next 7 days
- Expected: Displays upcoming classes

2. Bulk Enrollment (File)

- Input: User=Teacher, File=Valid XLSX (emails.xlsx)
- Expected: Students enrolled